TT-P-52L
March 6, 1973
SUPERSEDING
Int. Fed. Spec. TT-P-CC52C(GSA-FSS)
August -, 1971 and
Fed. Spec. TT-P-52L
April 16, 1965

FEDERAL SPECIFICATION

PAINT, OIL (ALKYD-OIL) WOOD SHAKES AND ROUGH SIDING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a relatively thin body paint for use on wood shakes, also suitable for painting rough siding and similar surfaces.

1.2 Classification.

1.2.1 Types. The ready-mixed paint shall be of four types.

Type I - White.
Type II - Light Tints.
Type III - Medium Tints.
Type IV - Mass or deep tones.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-L-11555 - Linseed Oil, Alkali Refined.
TT-P-143 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.
TT-T-291 - Thinner; Paint, Volatile Spirits (Petroleum-Spirits).

Federal Standards:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials (ASTM) Standards:

D 476 - Titanium Dioxide Pigments.
D 2088 - Determination of Low Concentration of Lead in Paint.
D 2206 - Determination of Low Quantities of Mercury in Paint.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The paint as received shall consist of pigments and vehicle specified, so combined as to produce a paint meeting all the requirements specified herein. The paint shall not contain lead in excess of 0.5 percent by weight of total nonvolatile matter calculated as lead metal.

3.1.1 Pigments. The pigments shall consist of prime pigment, extenders and suitable tinting pigments to produce the color desired (see table I and 6.2). 1/

3.1.2 Vehicle. The vehicle shall consist of a blend of alkali refined linseed oil conforming to TT-L-1155, alkyd resin, and suitable drier. The thinner or solvent used in the process shall be mineral spirits conforming to TT-T-291, type II or any solvent system complying with Air Pollution Regulations, "Rule 66". 2/

3.1.3 Color. The color of the paint specified in the contract or purchase order (see 6.2) shall match the standard color chip in Fed. Std. No. 595 (see table II).

3.2 Qualitative requirements.

3.2.1 Condition in container. The paint shall be thoroughly ground and mixed and shall not settle, cake, or thicken to such a degree that it cannot be easily redispersed with a paddle to a good brushing consistency when tested as in 4.3.1.

3.2.2 Storage stability.

3.2.2.1 Partially full container. A three-quarter filled, closed 8-ounce glass jar of paint shall show no skinning when tested as in 4.3.2.1. After aging (see 4.3.2.1) the paint shall show no livering, curdling, seeding, hard caking, or gummy sediment. It shall mix readily to a smooth honogeneous state and any skin formed shall be continuous and easily removed.

3.2.2.2 Full container. A full quart container of paint when tested as in 4.3.2.2 shall show no skinning, livering, curdling, seeding, hard, dry caking nor tough gummy sediment. At the end of one year storage (see 4.3.2.2) the paint shall be useable. It shall remix readily to a smooth homogeneous state, shall have a maximum consistency of 85 K.U. and shall meet all other requirements of the specification.

3.2.3 Dilution stability. The paint shall remain stable and uniform showing no precipitation, curdling or separation when reduced in the proportion of 8 parts by volume of paint and 1 part by volume of mineral spirits conforming to TT-T-291, type II when tested as in 4.3.3.

3.2.4 Odor. The odor of the freshly opened can and during application shall not be abnormally pungent, offensive, disagreeable or other than the normal characteristic of the paint when tested as in 4.3.4.

3.2.5 Application, spreading, and leveling properties. The paint shall show easy brushing, good leveling, and spreading when tested as in 4.3.5. In addition, it shall have no tendency to fill indentations or any tendency to sag off the end of the shakes or to leave the striation points bare.

3.2.6 Flexibility. A film of the paint when tested as in 4.3.6 shall withstand bending without cracking or flaking.

1/ The prime pigment for white shall be titanium dioxide conforming to ASTM D 476 type III or IV.
2/ Information on Rule 66 may be obtained from Los Angeles Air Pollution Control District, Los Angeles, California 90013.

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3.2.7 Knife test. A film of the paint tested as in 4.3.7 shall be hard and tough and shall adhere tightly to the metal. It shall be difficult to furrow off with the knife and shall not flake, chip, or powder. The knifed edge shall show beveled edges.

3.2.8 Water resistance. A film of the paint tested as in 4.3.8 shall show no blistering or wrinkling immediately after removal of panel from the water. The paint shall be no more than slightly affected when examined 2 hours after removal. After 24 hours air drying, the portion of the panel which was immersed shall be almost indistinguishable with regard to hardness, adhesion and general appearance from panel prepared at the same time but not immersed.

3.2.9 Fungus resistance. The paint shall show no fungus growth when tested as in 4.3.9. Mercurial type fungicide shall not be used. A certified statement from the supplier to this effect is necessary.

3.3 Quantitative requirements. The paint shall meet the quantitative requirements specified in table I. For pigmentation for specific colors see table II.

TABLE I. Quantitative requirements

Characteristics	Requirements							
	Type I		Type II		Type III		Type IV	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Reflectance	----	----	70	----	55	59	----	14
Weight per gallon (pounds)	11.0	----	11.0	----	11.0	----	10.0	----
Pigment (percent by weight of paint)	47	53	47	53	47	53	40	----
Total solids (percent by weight of paint)	70	----	70	----	70	----	62	----
Vehicle solids (percent by weight of vehicle)	50	----	50	----	50	----	40	----
Absorption (in inches)	----	3/4	----	3/4	----	3/4	----	3/4
Consistency (Kerb units)	72	82	69	75	69	75	69	75
Drying time:								
Set to touch (hours)	----	3	----	3	----	3	----	3
Dry hard (hours)	----	18	----	18	----	18	----	18
Coarse particles (percent by weight of paint)	----	2	----	2	----	2	----	2
Fineness of grind	4	----	4	----	4	----	4	----
Gloss 60° specular	----	6	----	6	----	6	----	6
Hiding power (contrast ratio (7 ml./square feet)	0.95	----	0.99	----	0.99	----	0.99	----
Phthalic anhydride (percent by weight of non volatile vehicle	12	----	12	----	12	----	12	----
Pigment composition:								
TiO_2 (anatase 20 percent, rutile 80 percent) (percent by weight of pigment)	42.5		----		----		----	----
TiO_2 (rutile) (percent by weight of pigment)	----		42.5		----		----	----
ZnO (percent by weight of pigment)	10.0		10.0		10.0		----	----
Extender, siliceous (percent by weight of pigment)	----	47.5	----	47.5	----	47.5	----	*
Lead content percent by weight of total nonvolatile	----	0.5	----	0.5	----	0.5	----	0.5
Mercury content	Negative		Negative		Negative		Negative	

* Balance of pigments (shading pigments and extenders to match color)

TABLE II. Pigmentation

Color		Percent		Suitable pigments
Fed. Std. No. 595		Prime pigment minimum	Extender maximum	
37875	Type I	53	47	Titanium dioxide (20 percent anatase - 80 percent rutile), zinc oxide*
36307	Type III	50	50	Yellow iron oxide, lampblack, titanium dioxide, zinc oxide*
36173	Type III	50	50	Iron oxide, lampblack, titanium dioxide, zinc oxide*
36251	Type III	50	50	Iron oxide, lampblack, titanium dioxide, zinc oxide*
36306	Type III	50	50	Iron oxide, lampblack, titanium dioxide, zinc oxide*

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TABLE II. Pigmentation

Color		Percent		
Fed. Std. No. 595		Prime pigment minimum	Extender maximum	Suitable pigments
34524	Type III	50	50	Chromic oxide, iron oxide, titanium dioxide, zinc oxide*
34226	Type III	50	50	Chromic oxide, lampblack, titanium dioxide, zinc oxide*
34325	Type III	50	50	Hansa yellow, iron oxide, phthalocyanine green, titanium dioxide, zinc oxide*
33695	Type II	50	50	Hansa yellow, iron oxide, titanium dioxide, zinc oxide*
30252	Type III	50	50	Iron oxide, Hansa yellow, dinitroaniline orange, titanium dioxide, zinc oxide*
30279	Type III	50	50	Iron oxide, titanium dioxide, zinc oxide*
34108	Type IV	42	50	Chromic oxide, phthalocyanine blue*
30045	Type IV	55	45	Brown iron oxide*
30111	Type IV	55	45	Red iron oxide*

* All extenders must be of the siliceous type. Percent prime pigments refers to only the active portion of a composite pigment; the extender portion must be included in the percent extenders. Trace amounts of other tinting colors necessary to obtain the proper shades are permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Production inspection of the paint.
(b) Inspection of preparation for delivery.

4.2.1 Sampling and inspection. Sampling and inspection shall be in accordance with method 1031 of Fed. Test Method Std. No. 141.

4.3 Test procedures. The paint shall be tested in accordance with Fed. Test Method Std. No. 141 as indicated in Table III except as otherwise specified.

TABLE III. Index

Characteristics	Requirement Reference	Applicable Methods	
		Fed. Test Method Std. No. 141	Paragraph Reference
Color	3.1.3	4250	-----
Condition in container	3.2.1	3011.1	-----
Storage stability	3.2.2	-----	4.3.2
Dilution stability	3.2.3	-----	4.3.3
Odor	3.2.4	-----	4.3.4
Application properties	3.2.5	-----	4.3.5
Flexibility	3.2.6	-----	4.3.6
Knife test	3.2.7	-----	4.3.7
Water resistance	3.2.8	-----	4.3.8
Gloss	Table I	6101	4.3.9
Reflectance	Table I	6121	4.3.9
Weight per gallon	Table I	4184.1	-----
Pigment	Table I	4022, 2021.1	-----
Vehicle solids	Table I	4052	-----
Total solids	Table I	4041.1	-----
Absorption	Table I	4421	-----
Fungus resistance	3.2.9	6271	4.3.10

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TABLE III. Index (con.)

Characteristics	Requirement Reference	Applicable Methods Fed. Test Method Std. No. 141	Paragraph Reference
Consistency	Table I	4281	-----
Drying time	Table I	4061.1	-----
Coarse particles	Table I	4092.1	-----
Fineness of grind	Table I	4411.1	-----
Hiding power (Contrast ratio)	Table I	4122.1	-----
Phthalic anhydride	Table I	7014	-----
Titanium dioxide (TiO_2)	Table I	7082	-----
Zinc oxide (ZnO)	Table I	7331	-----
Extenders	Table I	7331	-----
Lead content	Table I	-----	4.3.11

4.3.1 Condition in container. Determine package condition of the paint in accordance with method 3011.1 of Federal Test Method Std. 141 and observe for compliance with 3.2.1.

4.3.2 Storage stability

4.3.2.1 Partially full container. Determine skinning after 48 hours in accordance with method 3021 of Fed. Test Method Std. No. 141 for compliance with 3.2.2.1. Reseal and age for 7 days at 60°C then observe for compliance with 3.2.2.1.

4.3.2.2 Full container. In accordance with method 3022 of Fed. Test Method Std. No. 141 allow a full standard quart can of the paint to stand undisturbed for 1 year and then examine the contents. Evaluate pigment settling or caking as in 4.3.1 but agitate the can for 5 minutes on the paint shaker prior to re-examination. Run consistency and make other applicable test to determine compliance with 3.2.2.2.

4.3.3 Dilution stability. Reduce 8 parts by volume of mineral spirits conforming to TT-T-291, type II to 1 part by volume of paint for compliance with 3.2.3.

4.3.4 Odor. Determine the odor of the paint in accordance with method 4401 of Fed. Test Method Std. No. 141 for compliance with 3.2.4.

4.3.5 Application properties. Using a 4-inch brush, apply the paint at a spreading rate of about 200 square feet per gallon approximately 4 square feet of unprimed wooden shakes (five 6- by 18- inch machine striated vertical shakes) vertically mounted. Note whether the paint brushes easily and whether it flows readily into the shake identations. Allow to dry for 18 hours at room temperature and observe whether the paint has dried to a uniform film free from shinners and whether any sagging or dripping has occured off the end of the shakes. Also observe whether the paint has uniformly adhered to the points of striations.

4.3.6 Flexibility. Determine flexibility in accordance with method 6221 of Fed. Test Method Std. No. 141. Apply a 2-inch wide film of paint with a film applicator that will deposit a dry film thickness of 0.0009 to 0.0011-inch on a smooth, clean, electrolytic plate tin panel prepared in accordance with method 2012 of Fed. Test Method Std. No. 141. Air dry for 18 hours at room temperature, bake for 5 hours at 105° ± 2°C. and cool for one-half hour at 23° ± 1.1°C. Bend over one-half inch mandrel, note for compliance with 3.2.6.

4.3.7 Knife test. Perform the knife test on the flat portion of the flexibility test panel (see 4.3.6) in accordance with method 6304 of Fed. Test Method Std. No. 141, and observe the results for compliance with 3.2.7.

4.3.8 Water resistance. Draw down a film of the paint with a 0.0020-inch (0.0040-inch gap clearance) film applicator on a tin panel prepared in accordance with method 2012 of Fed. Test Method Std. No. 141 using the aliphatic naptha ethylene glycol monoethyl ether mixture. Air dry the paint for 96 hours. Then immerse in distilled water at 23° ± 1°C. for 18 hours in accordance with method 6011 of Fed. Test Method Std. No. 141. At the end of the test period remove and examine for compliance with 3.2.8.

4.3.9 Gloss.

4.3.10 Fungus resistance. Determine fungus or mildew resistance in accordance with method 6271 of Fed. Test Method Std. No. 141 for compliance with 3.2.9. Determination of mercury content as fungicide shall be in accordance with ASTM D2206.

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4.3.11 Lead content. Determine lead content in accordance with ASTM D2088 for compliance with 3.1 and Table I.

4.3.11.1 Calculation:

$$\text{Lead (percent by weight of total nonvolatile} = \frac{A \times 0.86623 \times B}{C \times W} \times 100$$

Where: A = Grams of lead oxide (PbO_2) in ash.

B = Percent ash by weight of paint divided by 100.

C = Percent total nonvolatile of paint divided by 100.

W = Grams of ash used.

4.4 Inspection of preparation for delivery. The paint shall be examined for compliance with packaging, packing, and marking requirements of section 5 in accordance with TT-P-143.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The paint shall be packaged, packed, and marked in accordance with TT-P-143. The level of packaging shall be A, B, or C, and the level of packing shall be A, B, C, as specified (see 6.2). The paint shall be furnished in 1-quart or 1-gallon multiple friction top containers, in 5-gallon lug cover steel pails or in 55-gallon steel drums as specified (see 6.2).

5.1.1 Precautionary marking. In addition to the markings required by TT-P-143, all individual containers containing volatile solvent shall have the following marking:

CAUTION: This paint contains volatile solvent. Adequate precautions should be taken when spraying.

6. NOTES

6.1 Intended use. The paint covered by this specification is intended for exterior use on wood shakes, rough siding and similar surfaces. The surface shall be cleaned of loose dirt and oil or grease before application of the paint.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and color required (see 1.2).
(c) Level of packaging and level of packing required (see section 5).
(d) Size of container required (see section 5).

6.3 The paint covered by this specification should be purchased by volume, the unit being one U. S. gallon of 231 cubic inches at 68°F. (20°C.).

6.4 The pigmentation contemplated in table I is the approximate combination of pigments for the specific colors.

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